2 **(a)** Gravitational force of Earth on satellite provides centripetal force on satellite.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$= \sqrt{\frac{\left(6.67\times10^{-11}\right)\left(6.0\times10^{24}\right)}{\left(6.4\times10^{6}+400\times10^{3}\right)}}$$

$$= 7671.6$$

$$= 7700 \text{ m s}^{-1}$$

(b)

$$U = -\frac{GMm}{r}$$

$$= -\frac{\left(6.67\times10^{-11}\right)\left(6.0\times10^{24}\right)(1200)}{\left(6.4\times10^{6}+400\times10^{3}\right)}$$

$$= -7.0624\times10^{10}$$

$$= -7.1\times10^{10} \text{ J}$$

(c) To just reach infinity, total energy of the spacecraft is always zero.

By principle of conservation of energy,

$$(\text{total energy})_{\text{at infinity}} = (\text{total energy})_{\text{at 400km}}$$

$$0 = \text{KE} + \left(-7.0624\times10^{10}\right)$$

$$\text{KE} = 7.0624\times10^{10} \text{ J}$$

$$\text{additional energy required} = 7.0624\times10^{10} - \frac{1}{2}(1200)(7671.6)^2$$

$$= 3.5311\times10^{10} = 3.5\times10^{10} \text{ J}$$

(d) **(i)** The kinetic energy of the spacecraft increases due to stronger gravitational acceleration at lower altitude while gravitational potential energy of the spacecraft decreases.

(ii) The total energy of the spacecraft at smaller orbital radius is more negative. Hence negative work has to be done to decrease the total energy.